

SIMATS ENGINEERING

CO1

ASSESSMENT TOOL 2

COURSE NAME: CSA07

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COURSE CODE: Computer Networks

REG NO: 192521064

1. Given

- **IP Address Block: 192.168.100.0/24**
- **School of Computing: /25**
- **School of Electronics: /26**
- **School of Mechanical Engineering: /27**

Subnet masks:

- **/25 = 255.255.255.128**
- **/26 = 255.255.255.192**
- **/27 = 255.255.255.224**

Step 1: Allocate the Subnets

1. School of Computing (/25)

- **Network Address: 192.168.100.0**
- **Subnet Mask: 255.255.255.128**
- **Broadcast Address: 192.168.100.127**
- **Valid Host Range: 192.168.100.1 – 192.168.100.126**
- **Usable Hosts: 126**

2. School of Electronics (/26)

- **Network Address: 192.168.100.128**
- **Subnet Mask: 255.255.255.192**
- **Broadcast Address: 192.168.100.191**
- **Valid Host Range: 192.168.100.129 – 192.168.100.190**
- **Usable Hosts: 62**

3. School of Mechanical Engineering (/27)

- **Network Address:** 192.168.100.192
- **Subnet Mask:** 255.255.255.224
- **Broadcast Address:** 192.168.100.223
- **Valid Host Range:** 192.168.100.193 – 192.168.100.222
- **Usable Hosts:** 30

Step 2: Remaining Unused IP Address Range

After allocating the above subnets:

- **Remaining Network:** 192.168.100.224/27
- **Network Address:** 192.168.100.224
- **Valid Host Range:** 192.168.100.225 – 192.168.100.254
- **Broadcast Address:** 192.168.100.255
- **Usable Hosts:** 30

Final Answer (Tabular Form)

Department	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.127	192.168.100.1 – 192.168.100.126	126
School of Electronics	/26	192.168.100.128	192.168.100.191	192.168.100.129 – 192.168.100.190	62
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.223	192.168.100.193 – 192.168.100.222	30
Unused IP Block	/27	192.168.100.224	192.168.100.255	192.168.100.225 – 192.168.100.254	30

Summary

- **School of Computing:** Network = **192.168.100.0/25**, Broadcast = **192.168.100.127**, Hosts = **126**
- **School of Electronics:** Network = **192.168.100.128/26**, Broadcast = **192.168.100.191**, Hosts = **62**
- **School of Mechanical Engineering:** Network = **192.168.100.192/27**, Broadcast = **192.168.100.223**, Hosts = **30**
- **Remaining Unused Block:** **192.168.100.224/27** with **30 usable host addresses**.

2. Given

- **Allocated IP Block: 10.10.0.0/16**
- **Software Development: /24**
- **Quality Assurance: /25**
- **Human Resources: /26**
- **Executive Management: /27**

Subnet Masks

- **/24 = 255.255.255.0**
- **/25 = 255.255.255.128**
- **/26 = 255.255.255.192**
- **/27 = 255.255.255.224**

Step 1: Allocate the Subnets

1. Software Development (/24)

- **Network Address: 10.10.0.0**
- **Subnet Mask: 255.255.255.0**
- **Broadcast Address: 10.10.0.255**
- **Valid Host Range: 10.10.0.1 – 10.10.0.254**
- **Usable Hosts: 254**

2. Quality Assurance (/25)

- **Network Address: 10.10.1.0**
- **Subnet Mask: 255.255.255.128**
- **Broadcast Address: 10.10.1.127**
- **Valid Host Range: 10.10.1.1 – 10.10.1.126**
- **Usable Hosts: 126**

3. Human Resources (/26)

- **Network Address: 10.10.1.128**
- **Subnet Mask: 255.255.255.192**
- **Broadcast Address: 10.10.1.191**
- **Valid Host Range: 10.10.1.129 – 10.10.1.190**
- **Usable Hosts: 62**

4. Executive Management (/27)

- **Network Address: 10.10.1.192**

- **Subnet Mask:** 255.255.255.224
- **Broadcast Address:** 10.10.1.223
- **Valid Host Range:** 10.10.1.193 – 10.10.1.222
- **Usable Hosts:** 30

Step 2: Remaining Unused IP Address Range

After allocating the above subnets, the remaining addresses in **10.10.1.x** are:

- **Unused Block:** 10.10.1.224/27
- **Network Address:** 10.10.1.224
- **Valid Host Range:** 10.10.1.225 – 10.10.1.254
- **Broadcast Address:** 10.10.1.255
- **Usable Hosts:** 30

Beyond that, since the original allocation is **10.10.0.0/16**, all addresses from **10.10.2.0** to **10.10.255.255** also remain unused.

Final Answer (Table)

Department	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance	/25	10.10.1.0	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources	/26	10.10.1.128	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management	/27	10.10.1.192	10.10.1.223	10.10.1.193 – 10.10.1.222	30

Remaining Unused IP Range

- **Within 10.10.1.x:** 10.10.1.224 – 10.10.1.255 (Subnet: 10.10.1.224/27)
- **Remaining part of the /16 block:** 10.10.2.0 – 10.10.255.255

This is the correct VLSM allocation for the **10.10.0.0/16** network.

3. Given

- **IPv6 Address:** 2001:0db8:abcd:0000:0000:0000:0000/64

1. Convert the IPv6 Address into Compressed Notation

Rules:

- Remove leading zeros in each block.
- Replace consecutive blocks of 0000 with :: (only once).

Compressed IPv6 Address:

2001:db8:abcd::/64

2. Expand the Compressed Address Back into Full Hexadecimal Representation

Expand :: back to the required number of 0000 blocks.

Expanded IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000/64

3. Identify the Network Prefix (/64)

With a /64 prefix:

- The **first 64 bits** represent the **network prefix**.
- The **last 64 bits** represent the **interface identifier (host portion)**.

Network Prefix:

2001:0db8:abcd:0000::/64

or simply

2001:db8:abcd:0::/64

4. Calculate the Total Number of Host Addresses Supported Within the Subnet

A /64 prefix leaves:

- **$128 - 64 = 64$ host bits**

Total host addresses:

$$[2^{\{64\}} = 18,446,744,073,709,551,616]$$

Total Host Addresses = 18,446,744,073,709,551,616

5. Total Number of Possible Host Addresses Available Within This Network

Since IPv6 does **not reserve network or broadcast addresses**, all ($2^{\{64\}}$) addresses are usable.

Possible Host Addresses = 18,446,744,073,709,551,616

Final Answer (Table)

Task	Answer
1. Compressed IPv6 Address	2001:db8:abcd::/64
2. Expanded IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000/64
3. Network Prefix (/64)	2001:db8:abcd:0::/64
4. Total Host Addresses Supported	$2^{64} = 18,446,744,073,709,551,616$
5. Total Possible Host Addresses	$18,446,744,073,709,551,616 (2^{64})$

Note: In IPv6, unlike IPv4, there is **no broadcast address**, and a /64 subnet provides 2^{64} possible interface addresses.

4. Given

Routing Table

Destination Network	Subnet Mask
192.168.1.0/24	255.255.255.0
192.168.2.0/24	255.255.255.0
192.168.3.0/24	255.255.255.0

Destination IP Address: 192.168.2.45

1. Determine the Destination Subnet

A /24 subnet means the first three octets represent the network.

- Destination IP: **192.168.2.45**
- Network Address: **192.168.2.0/24**

Answer: The packet belongs to the **192.168.2.0/24** subnet.

2. Matching Routing Table Entry

The router checks its routing table using the **Longest Prefix Match**.

The matching entry is:

192.168.2.0/24

3. Next-Hop Network/Interface

Since the destination belongs to **192.168.2.0/24**, the router forwards the packet through the **interface or next hop associated with the 192.168.2.0/24 route**.

Answer:

- **Next-hop Network: 192.168.2.0/24**
- **Outgoing Interface:** The interface configured for the **192.168.2.0/24** network (for example, **Interface G0/1**, depending on the router configuration).

Note: The exact interface name (such as G0/1 or Fa0/0) cannot be determined because it is not provided in the question.

4. Default Route

If the destination IP does **not** match any routing table entry, the router forwards the packet using the **default route**.

Default Route:

0.0.0.0/0

This route sends the packet to the default gateway (next-hop router).

Final Answer (Table)

Task	Answer
1. Destination Subnet	192.168.2.0/24
2. Matching Routing Table Entry	192.168.2.0/24
3. Next-Hop Network/Interface	192.168.2.0/24 (through the interface configured for this network, e.g., G0/1)
4. Default Route	0.0.0.0/0 (used when no route matches)

This answer is suitable for assignment and exam purposes.

5. Given

- **Administration LAN: 192.168.10.0/26**
- **Finance LAN: 192.168.10.64/26**

Router Interfaces (Gateways):

- **Administration Gateway: 192.168.10.1**
- **Finance Gateway: 192.168.10.65**

Subnet Mask (/26): 255.255.255.192

1. Determine the Network Address and Broadcast Address

Administration LAN

- **Network Address: 192.168.10.0**
- **Broadcast Address: 192.168.10.63**

Finance LAN

- **Network Address: 192.168.10.64**
- **Broadcast Address: 192.168.10.127**

2. Valid Host IP Address Range

Administration LAN

- **Host Range: 192.168.10.1 – 192.168.10.62**
- **Usable Hosts: 62**

Finance LAN

- **Host Range: 192.168.10.65 – 192.168.10.126**
- **Usable Hosts: 62**

3. Suitable IP Addresses for Three Computers

Administration LAN

Computer	IP Address
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PC1	192.168.10.2
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PC2	192.168.10.3
-----	--------------

PC3	192.168.10.4
-----	--------------

(Gateway: 192.168.10.1)

Finance LAN

Computer	IP Address
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PC1	192.168.10.66
-----	---------------

PC2	192.168.10.67
-----	---------------

PC3	192.168.10.68
-----	---------------

(Gateway: 192.168.10.65)

4. Default Gateway for Each LAN

LAN	Default Gateway
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Administration LAN	192.168.10.1
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Finance LAN	192.168.10.65
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Final Answer (Table)

Department	Network Address	Broadcast Address	Valid Host Range	Example Host IPs	Default Gateway
Administration LAN	192.168.10.0/26	192.168.10.63	192.168.10.1 – 192.168.10.62	192.168.10.2, 192.168.10.3, 192.168.10.4	192.168.10.1
Finance LAN	192.168.10.64/26	192.168.10.127	192.168.10.65 – 192.168.10.126	192.168.10.66, 192.168.10.67, 192.168.10.68	192.168.10.65

This is the correct subnetting and addressing plan for the two LANs and is suitable for assignment or exam submission.